

Large Language Models (LLMs)

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What are Large Language Models (LLMs)?

A Large Language Model is a Transformer-based neural network trained on enormous amounts of text so that it can understand and generate human language.

Why are They Called Large?

Because they have

- Huge datasets (books, websites, articles, code, research papers)
- Millions or billions of parameters
- Massive computational resources (GPUs/TPUs)
- Long training durations

Why Do LLMs Sound Intelligent?

They learn

- Grammar
- Vocabulary
- Facts (up to their training and available tools)
- Reasoning patterns
- Programming languages
- Writing styles
- Problem-solving strategies

They do **not** think like humans; they **predict the most appropriate next token** based on learned patterns and context.

Popular LLMs

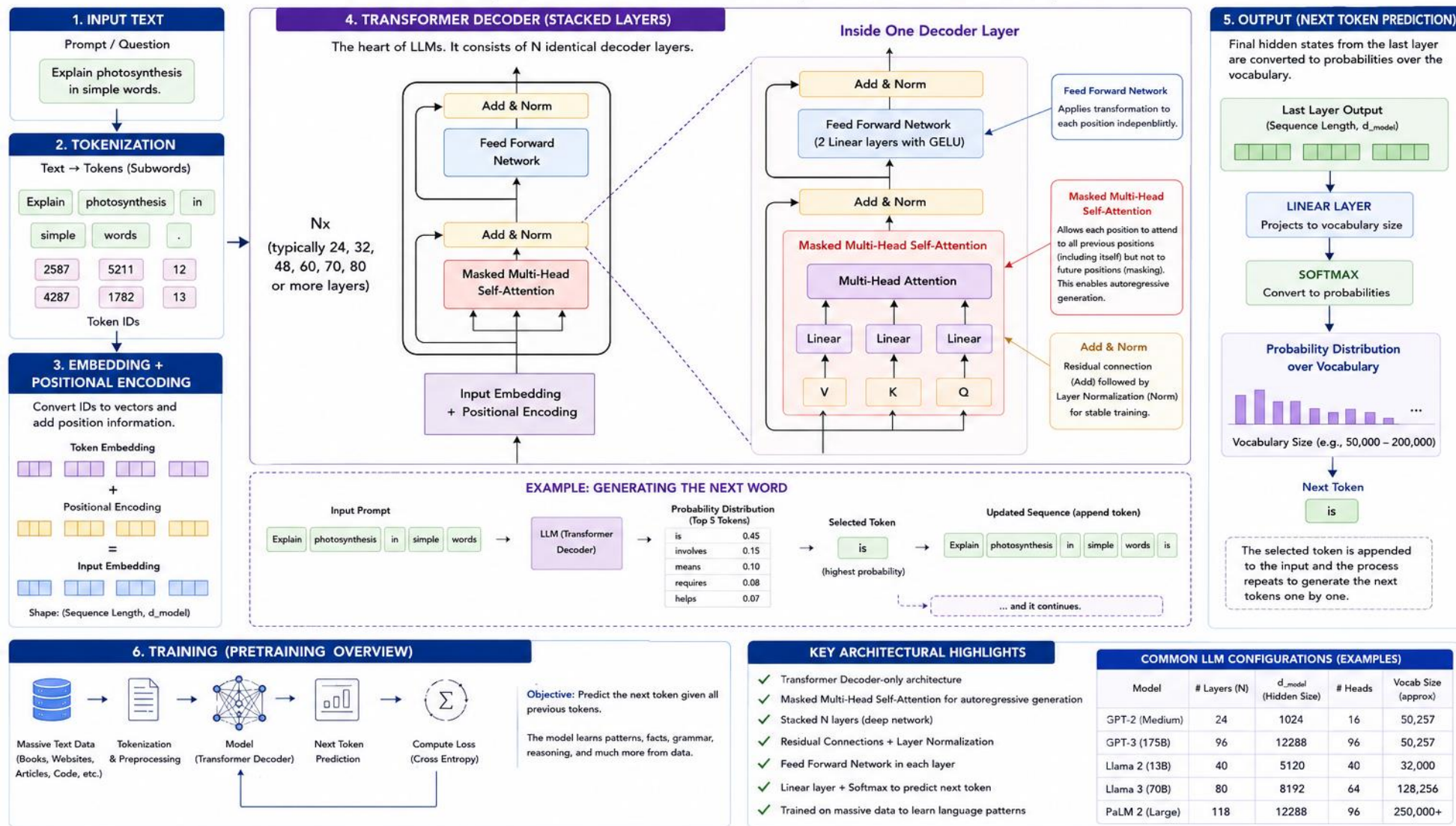
Model	Organization	Major Use
ChatGPT	OpenAI	General-purpose AI assistant
Gemini	Google	Search, multimodal AI
Claude	Anthropic	Long-document reasoning
Llama	Meta	Open-weight AI models

Applications of LLMs

- Intelligent Chatbots
- AI Coding Assistants
- Machine Translation
- Document Summarization
- Content Generation
- Question Answering
- Medical Report Analysis
- Legal Document Analysis
- Education and Tutoring
- Research Assistance

ARCHITECTURE OF A LARGE LANGUAGE MODEL (LLM)

An LLM is basically a Transformer Decoder model that predicts the next token in a sequence.



In One Line

An LLM takes a sequence of tokens, converts them to vectors, processes them through many Transformer decoder layers using masked self-attention, and predicts the most likely next token. This is repeated to generate human-like text.

- Thank you